

Radical telescopes for Chebyshev transfers of every degree: normalization, programmed squares, and the fixed-orbit boundary

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This is a non-peer-reviewed companion research note. It consolidates results developed after the quadratic and prime-degree Chebyshev-orbit manuscripts. The transfer is classical and compositional. The proposed contribution is the degree-uniform arithmetic package described below; specialist review of its proofs, sources, and priority remains necessary.

Abstract

For every integer $d \geq 2$, homogenizing T_d gives an integral map from a primitive positive triple $a + b = c$ of opposite parity to another such triple. We identify one fixed Lucas sequence behind the whole orbit. After a necessary normalization at the primes dividing d , the quotient between successive indices d^j is a positive integer $E_{d,j}$. Its cyclotomic indices are precisely

$$\{m : m \mid d^{j+1}, m \nmid d^j\}.$$

For a natural class of d -admissible seeds, the $E_{d,j}$ are pairwise coprime and disjoint from the seed. This yields the exact all-degree radical telescope

$$\frac{\text{rad}(a_n b_n c_n)}{c_n} = \frac{\text{rad}(a_0 b_0 c_0)}{c_0} \frac{|\sin(d^n \theta)|}{d^n \sin \theta W_{d,n}}, \quad W_{d,n} = \prod_{j < n} \frac{E_{d,j}}{\text{rad}(E_{d,j})}.$$

Composite degree bundles several Lucas atoms in one layer; prime degree is the single-atom case. We classify the coordinate atoms at every index, complete the simultaneous finite-genealogy realization for the quadratic transfer, and prove a bounded iterated local mean for the logarithmic defect in every degree. For one fixed orbit we isolate an unconditional growing congruence window and show that every fixed valuation depth in a polynomial window is harmless; the remaining obstruction is an exceptionally deep Lucas–Wieferich tail. Two independent implementations scan the canonical quadratic, cubic, and quintic pairs through 10^8 : four Lucas–Wieferich primes occur in total, none tower-compatible or super-Wieferich. A 122-digit quadratic layer is also completely factored. These results do not prove the fixed-orbit conjecture or any new case of the *abc* conjecture.

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Introduction and novelty boundary

Write

$$\text{rad}(N) = \prod_{p|N} p.$$

A primitive positive triple $a + b = c$ is an abc -hit when $\text{rad}(abc) < c$. Polynomial identities that transfer one additive triple to another are classical; Martin and Miao survey many examples, and van der Horst studies the transfer method in detail. The Chebyshev identity used here is a homogenized de Moivre identity, not a new polynomial identity.

Lucas atoms and their valuations have also been developed independently of this project. Sagan and Tirrell introduce the atom formalism, while Alecci, Miska, Murru, and Romeo give a general definition and detailed p -adic valuation results. The finite-field factorization behind the local root counts is equivalent, after a Cayley change of variable, to the fibotomic factorization of Byer, Dvorachek, Eckard, Harrington, Wise, and Wong. Ratliff, Rush, and Shah give related radical-preserving cyclotomic constructions.

The proposed contribution is therefore narrower than “a new Chebyshev map.” It is the following synthesis:

one normalization valid for every degree; prime-support separation of its successive multi-atom layers; an exact radical telescope along the resulting additive orbit; constructive control of arbitrary finite square patterns in every prime degree; an all-index coordinate-atom classification; and a bounded local first moment in every degree whose remaining integer-box obstruction is stated explicitly.

Prime degree remains special: a new layer is a single cyclotomic atom (or two coordinate branches of the same order), whereas composite degree groups several atom indices. Since $T_m \circ T_n = T_{mn}$, the composite-degree transfer is itself only a block of lower-degree transfers. The grouping and normalization, rather than the map, are the point.

The last sections distinguish finite-seed averages from a single fixed orbit. They sharpen the fixed-orbit obstruction but do not resolve it. In particular, nothing here should be read as a proof of abc .

The universal integral transfer

Let $a + b = c$ be primitive and positive, with a, b of opposite parity, and put

$$D = b - a.$$

Both D and c are odd. Define the homogeneous Chebyshev values

$$H_r(D, c) = c^r T_r(D/c).$$

They are integral because

$$H_0 = 1, \quad H_1 = D, \quad H_{r+1} = 2DH_r - c^2 H_{r-1}.$$

Theorem 1 (Universal transfer). *For every integer $d \geq 2$, set*

$$\mathcal{T}_d(a, b, c) = \left(\frac{c^d - H_d(D, c)}{2}, \frac{c^d + H_d(D, c)}{2}, c^d \right). \quad (2.1)$$

Then $\mathcal{T}_d$ maps primitive positive opposite-parity triples to triples of the same kind. If

$$x = \frac{b - a}{c},$$

then the new normalized difference is $\mathcal{T}_d(x)$. Moreover

$$\mathcal{T}_m \circ \mathcal{T}_n = \mathcal{T}_{mn} = \mathcal{T}_n \circ \mathcal{T}_m. \quad (2.2)$$

After n iterations,

$$c_n = c_0^{d^n}, \quad \frac{b_n - a_n}{c_n} = \mathcal{T}_{d^n} \left(\frac{b_0 - a_0}{c_0} \right). \quad (2.3)$$

Proof. Since $H_d(D, c)$ is odd, the two coordinates in (2.1) are integers of opposite parity and sum to c^d . The inequality $|\mathcal{T}_d(x)| \leq 1$ gives nonnegativity. Equality would make $e^{i \arccos x}$ a root of unity, so $2D/c$ would be a rational algebraic integer. This is impossible because $c > 1$, c is odd, and $\gcd(D, c) = 1$. Thus both coordinates are positive.

If an odd prime divided both new coordinates, it would divide c and $H_d(D, c)$. But

$$H_d(D, c) \equiv 2^{d-1} D^d \pmod{c},$$

which is a unit modulo every prime divisor of c . The prime 2 cannot divide both coordinates because their sum is odd. This proves primitivity. The difference of the two new coordinates is $H_d(D, c)$, which proves the semiconjugacy. Equations (2.2) and (2.3) follow from $\mathcal{T}_m \circ \mathcal{T}_n = \mathcal{T}_{mn}$. $\square$

For $d = 2$, (2.1) is exactly

$$(a, b, c) \mapsto (4ab, (b - a)^2, c^2). \quad (2.4)$$

For odd d , expanding $(\sqrt{b} + i\sqrt{a})^d$ splits the transfer into the familiar two coordinate factors. For even d , write instead

$$(\sqrt{b} + i\sqrt{a})^d = C_d(a, b) + i\sqrt{ab} S_d(a, b).$$

Then

$$\mathcal{T}_d(a, b) = \begin{cases} (aS_d(a, b)^2, bC_d(a, b)^2), & d \text{ odd}, \\ (abS_d(a, b)^2, C_d(a, b)^2), & d \text{ even}. \end{cases} \quad (2.5)$$

Thus every degree has a two-coordinate-factor form, although the seed prefactor depends on parity.

One Lucas sequence and its normalized layers

Fix the seed and put

$$\alpha = D + 2\sqrt{-a_0 b_0}, \quad \beta = D - 2\sqrt{-a_0 b_0}.$$

Then

$$\alpha + \beta = 2D, \quad \alpha\beta = c_0^2, \quad (\alpha - \beta)^2 = -16a_0 b_0.$$

The associated Lucas sequence is

$$U_0 = 0, \quad U_1 = 1, \quad U_{r+1} = 2DU_r - c_0^2 U_{r-1}, \quad U_r = \frac{\alpha^r - \beta^r}{\alpha - \beta}. \quad (3.1)$$

Lemma 2 (Fixed-sequence identity). *For the degree- d orbit,*

$$a_n b_n = a_0 b_0 U_d^2. \quad (3.2)$$

Proof. The norm identity $c_0^{2d^n} - (\alpha^{d^n} + \beta^{d^n})^2/4 = -(\alpha^{d^n} - \beta^{d^n})^2/4$, together with $(\alpha - \beta)^2 = -16a_0b_0$, gives (3.2). $\square$

Define

$$Q_{d,j} = \frac{U_{d^{j+1}}}{U_{d^j}}. \quad (3.3)$$

Homogeneous cyclotomic factorization gives

$$Q_{d,j} = \prod_{m \in \mathcal{L}_{d,j}} \Phi_m(\alpha, \beta), \quad \mathcal{L}_{d,j} = \{m : m \mid d^{j+1}, m \nmid d^j\}. \quad (3.4)$$

The sets $\mathcal{L}_{d,j}$ are pairwise disjoint.

The quotient (3.3) contains a systematic contribution from every prime dividing d . The following seed condition removes it exactly.

Definition 3. A seed is *d-admissible* when it is primitive, positive, and of opposite parity, and for every odd prime $q \mid d$,

$$q \mid a_0, \quad v_q(U_q) = 1. \quad (3.5)$$

For $q \geq 5$, the second condition follows automatically from the first. At $q = 3$ it is necessary: the end terms in the binomial expression for U_q/q can cancel modulo q .

Theorem 4 (Normalized layer genealogy). *Let $d \geq 2$ and let the seed be d-admissible. Then*

$$E_{d,j} = \frac{|Q_{d,j}|}{d} \quad (3.6)$$

is a positive integer coprime to $da_0b_0c_0$. The integers

$$E_{d,0}, E_{d,1}, E_{d,2}, \dots$$

are pairwise coprime. Consequently every prime outside the seed support has at most one birth layer.

Proof. For an odd $q \mid d$, one has $q \mid (\alpha - \beta)^2$ because $q \mid a_0$. The Lucas law of repetition at this discriminant prime, under (3.5), gives

$$v_q(U_r) = v_q(r) \quad (q \mid r). \quad (3.7)$$

Opposite parity gives the corresponding identity

$$v_2(U_r) = v_2(r) \quad (2 \mid r). \quad (3.8)$$

These are standard specializations of Lucas-sequence valuation formulas; the hypotheses include $q \nmid \alpha\beta$ and nondegeneracy. Subtracting the valuations at d^{j+1} and d^j gives

$$v_q(Q_{d,j}) = v_q(d) \quad (q \mid d),$$

so (3.6) is integral and coprime to d .

If an odd $p \mid a_0 b_0$, then $U_r \equiv r\alpha^{r-1} \pmod{p}$. This is nonzero when $p \nmid d$, and (3.7) removes the exact systematic part when $p \mid d$. If $p \mid c_0$, one of α, β is zero and the other is a unit modulo p . The parity assertion follows from (3.8) when d is even and directly from the recurrence at odd indices when d is odd. Hence the normalized layers avoid the seed support.

Now suppose $p \nmid da_0 b_0 c_0$ divided two layers. At a prime above p , a factor $\Phi_m(\alpha, \beta)$ with $p \nmid m$ says that α/β has exact order m ; inversion under conjugation does not change that order. The same p cannot therefore occur at indices from two disjoint sets $\mathcal{L}_{d,j}$. This proves pairwise coprimality. $\square$

Remark 5. For an odd prime $d = \ell$, a layer is the single prime-power cyclotomic index ℓ^{j+1} , expressed through two coordinate branches. For $d = 2$, (3.6) is $|b_j - a_j|$. A composite d bundles several atoms in (3.4). Formula (3.6), not a termwise composite-binomial argument, is the stable normalization.

The all-degree radical telescope

Let

$$R_n = \text{rad}(a_n b_n c_n), \quad W_{d,n} = \prod_{j < n} \frac{E_{d,j}}{\text{rad}(E_{d,j})}. \quad (4.1)$$

Every prime divisor of d already lies in the seed radical: the odd ones divide a_0 , and opposite parity handles 2. Telescoping (3.3) and using Theorem 4 therefore gives

$$|U_{d^n}| = d^n \prod_{j < n} E_{d,j}, \quad R_n = R_0 \prod_{j < n} \text{rad}(E_{d,j}). \quad (4.2)$$

Choose $\theta \in (0, \pi)$ by

$$\cos \theta = \frac{b_0 - a_0}{c_0}.$$

Since $\alpha = c_0 e^{i\theta}$,

$$U_{d^n} = c_0^{d^n - 1} \frac{\sin(d^n \theta)}{\sin \theta}. \quad (4.3)$$

Theorem 6 (All-degree radical telescope). *For every d -admissible seed and every $n \geq 0$,*

$$\boxed{\frac{R_n}{c_n} = \frac{R_0}{c_0} \frac{|\sin(d^n \theta)|}{d^n \sin \theta W_{d,n}}}. \quad (4.4)$$

Proof. Divide the two identities in (4.2), substitute (4.3), and use $c_n = c_0^{d^n}$. $\square$

For a fixed algebraic seed, a standard linear-form estimate supplies a constant $C = C(a_0, b_0, d)$ such that $\log |\sin(d^n \theta)| \geq -C(n+1)$; the elementary upper bound is immediate. Thus the exponential-scale obstruction in (4.4) is precisely the powerful part $W_{d,n}$. In the quadratic and prime-degree cases this yields the familiar equivalence

$$\log W_{d,n} = o(d^n) \iff \text{the orbit quality tends to 1.} \quad (4.5)$$

A persistent quality gap would require $\log W_{d,n}$ to have positive scale d^n . Equation (4.4) is an exact reduction, not a bound on that powerful part.

Programmed squares in prime degree

The prime-degree manuscript proves that any finite compatible collection of birth levels, branches, splitting signs, and exact multiplicities can be realized simultaneously for every odd prime degree. Here we record the missing quadratic counterpart. It is also useful as a source of exact test vectors for the fixed-orbit discussion.

For the quadratic transfer (2.4), let

$$G_n(X, Y) = b_n - a_n$$

be the homogeneous seed-coordinate form and put $m = 2^{n+2}$.

Proposition 7 (Quadratic local roots). *Let p be odd. If $m \mid p - \chi$ for $\chi \in \{1, -1\}$, then $G_n(X, 1)$ has exactly 2^n distinct simple roots in $\mathbb{F}_p$. They are*

$$\rho_\zeta = - \left(\frac{\zeta - 1}{\zeta + 1} \right)^2, \quad (5.1)$$

where ζ has exact order m in $\mathbb{F}_p^\times$ when $\chi = 1$, or in the norm-one subgroup of $\mathbb{F}_{p^2}^\times$ when $\chi = -1$, modulo $\zeta \sim \zeta^{-1}$. There are no roots when neither divisibility holds. Every root satisfies

$$\rho \neq 0, -1, \quad \left(\frac{-\rho}{p} \right) = \chi. \quad (5.2)$$

Each root has a unique Hensel lift. Let $\hat{p}$ be an integer representing the p -adic Hensel root modulo p^{h+1} . Then for every unit λ ,

$$v_p(G_n(\hat{p} + \lambda p^h, 1)) = h. \quad (5.3)$$

Proof. In the quadratic extension generated by square roots of X and Y , the standard Cayley parameter identifies the roots of G_n with exact order- m roots of unity, and inversion has the same seed-coordinate image. The split and norm-one groups have orders $p-1$ and $p+1$, respectively. The 2^n distinct constructed roots exhaust the degree- 2^n polynomial $G_n(X, 1)$, whose leading coefficient is a unit; hence all roots are simple. The exceptional values $0, -1$ are not roots. Formula (5.2) follows directly from (5.1). Hensel's lemma and the first nonzero Taylor term give (5.3). $\square$

Theorem 8 (Simultaneous quadratic realization). *Let*

$$\mathcal{D} = \{(p_i, n_i, h_i, \chi_i) : 1 \leq i \leq r\}$$

have distinct odd primes, $n_i \geq 0$, $h_i \geq 1$, and

$$2^{n_i+2} \mid p_i - \chi_i.$$

There are infinitely many primitive positive seeds with a_0 even and b_0 odd such that, simultaneously for every i ,

$$v_{p_i}(E_{2, n_i}) = h_i, \quad \left(\frac{D_K}{p_i} \right) = \chi_i, \quad (5.4)$$

where $K = \mathbb{Q}(\sqrt{-a_0 b_0})$, while p_i divides neither the seed nor any layer $E_{2, j}$ with $j \neq n_i$.

Proof. For each datum choose a root from Proposition 7 and prescribe its Hensel lift with the perturbation in (5.3) modulo $p_i^{h_i+1}$. Fix $b_0 = 1$. The Chinese remainder theorem, together with $a_0 \equiv 0 \pmod{2}$, supplies infinitely many positive a_0 in the resulting progression. These seeds are primitive and have the required parity. Equations (5.2) and (5.3) give the splitting sign and exact valuation. The exclusions $\rho \neq 0, -1$ keep p_i away from $a_0 b_0 c_0$. Finally, the Cayley parameter has exact order 2^{n_i+2} , so it cannot belong to the distinct order class attached to another layer. This proves the last assertion. $\square$

Theorem 8, together with the two-branch odd-prime result, shows that arbitrarily deep but finitely prescribed square behavior is constructible. It does not produce a fixed seed with large defect at infinitely many layers.

A bounded local mean for every degree

This section concerns seed averages, not one orbit. For any $d \geq 2$, let $\mathcal{S}_d(H)$ be the d -admissible seed pairs in $[1, H]^2$. The defining conditions occur at the fixed primes dividing $2d$. Möbius inversion and the Chinese remainder theorem give

$$|\mathcal{S}_d(H)| \sim \kappa_d H^2 \quad (H \rightarrow \infty) \quad (6.1)$$

for an effective $\kappa_d > 0$. At 3, nonemptiness of the valuation-one condition follows, for example, from $a \equiv 3, b \equiv 2 \pmod{9}$; the conditions at the other odd prime divisors of d are automatic once they divide a .

We first remove the prime-degree restriction from the local root count. In the universal algebra

$$\mathcal{R} = \mathbb{Z}[X, Y, r, t]/(r^2 - Y, t^2 + X), \quad \Omega = r + t, \quad \bar{\Omega} = r - t, \quad (6.2)$$

define, for every $k \geq 3$,

$$A_k(X, Y) = \Phi_k(\Omega, \bar{\Omega}), \quad (6.3)$$

where $\Phi_k(U, V)$ is the homogeneous cyclotomic polynomial.

Lemma 9 (All-index coordinate atoms). *The expression in (6.3) belongs to $\mathbb{Z}[X, Y]$, is homogeneous of degree $\varphi(k)/2$, and has the following roots. Let $p \nmid 2k$ be prime. The polynomial $A_k(X, 1)$ has roots over $\mathbb{F}_p$ precisely when*

$$k \mid p - \chi \quad \text{for one } \chi \in \{1, -1\}. \quad (6.4)$$

When (6.4) holds, it has exactly $\varphi(k)/2$ distinct simple roots

$$\rho_\zeta = - \left(\frac{\zeta - 1}{\zeta + 1} \right)^2, \quad (6.5)$$

where ζ , modulo $\zeta \sim \zeta^{-1}$, ranges over the elements of exact order k in $\mathbb{F}_p^\times$ for $\chi = 1$, or in the norm-one subgroup of $\mathbb{F}_{p^2}^\times$ for $\chi = -1$. Every root satisfies

$$\rho_\zeta \neq 0, -1, \quad \left(\frac{-\rho_\zeta}{p} \right) = \chi. \quad (6.6)$$

For $p \nmid kk'$, the root sets of distinct indices $k, k' \geq 3$ are disjoint. If $\hat{\rho}_{h+1}$ represents the Hensel lift of a root modulo p^{h+1} , then for every $h \geq 1$ and every p -adic unit u ,

$$v_p(A_k(\hat{\rho}_{h+1} + up^h, 1)) = h. \quad (6.7)$$

Proof. For $k > 2$, the polynomial $\Phi_k(U, V)$ is symmetric of even total degree $\varphi(k)$. In its integral expression in the elementary symmetric functions, homogeneity forces the exponent of $e_1 = U + V$ to be even. Substituting

$$e_1^2 = 4Y, \quad e_2 = UV = X + Y$$

proves the integrality and degree assertion.

If (6.4) holds, choose ζ of exact order k in the indicated cyclic group and put

$$s = \frac{\zeta - 1}{\zeta + 1}, \quad \rho = -s^2.$$

In the split case s lies in $\mathbb{F}_p$; in the norm-one case $s^p = -s$. Thus $\rho \in \mathbb{F}_p$, and at $r = 1, t = s$ one has $\Omega/\overline{\Omega} = \zeta$, so (6.3) vanishes. The map from ζ to ρ has exactly the inversion fibers, giving $\varphi(k)/2$ distinct roots and (6.6).

Conversely, a root is neither 0 nor -1 : these specializations give quotient 1 or make one of $\Omega, \overline{\Omega}$ zero, and the corresponding homogeneous cyclotomic value is a unit modulo $p \nmid 2k$. Taking $s^2 = -\rho$ recovers an exact-order- k Cayley parameter in the split or norm-one group, proving (6.4) and exhaustion. The leading coefficient is also a unit modulo p : at $Y = 0$ the quotient is -1 , and any prime divisor of the resulting cyclotomic value divides $2k$. Hence the roots are simple. A common root at two indices would give the same Cayley parameter, up to inversion, two different exact orders. Finally, Hensel's lemma and the first nonzero Taylor term modulo p^{h+1} give (6.7). $\square$

Put

$$\Lambda_{d,j} = \{k : k \mid 2d^{j+1}, k \nmid 2d^j\}. \quad (6.8)$$

Since $\alpha = \Omega^2, \beta = \overline{\Omega}^2$, homogeneous cyclotomic factorization gives the uniform coordinate decomposition

$$U_N = \prod_{\substack{k \mid 2N \\ k > 2}} A_k(a, b), \quad Q_{d,j} = \prod_{k \in \Lambda_{d,j}} A_k(a, b). \quad (6.9)$$

The sets $\Lambda_{d,j}$ are pairwise disjoint. In particular, $\Lambda_{2,j} = \{2^{j+2}\}$; for odd d it is the union of the odd Lucas indices in $\mathcal{L}_{d,j}$ and their doubles. Division by the systematic content d in (3.6) changes neither roots nor valuations at the primes $p \nmid 2d$ considered below.

For fixed n, P, K , define the bounded defect

$$D_{d,n,P,K}(a, b) = \sum_{j < n} \sum_{\substack{p \leq P \\ p \nmid 2d}} \log p \sum_{h=2}^{K+1} \mathbf{1}_{p^h \mid E_{d,j}(a,b)}. \quad (6.10)$$

For a Lucas index $m \geq 2$, put

$$\lambda(m) = \begin{cases} m, & m \text{ odd}, \\ 2m, & m \text{ even}. \end{cases} \quad (6.11)$$

The identity

$$\Phi_m(\alpha, \beta) = \begin{cases} A_m(a, b) A_{2m}(a, b), & m \text{ odd}, \\ A_{2m}(a, b), & m \text{ even} \end{cases} \quad (6.12)$$

shows that the coordinate-index and Lucas-index descriptions below are equivalent. For odd m , the compatibility conditions for A_m and A_{2m} coincide: if $m \mid p - \chi$, then $2m \mid p - \chi$ because $p - \chi$ is even.

Theorem 10 (All-degree bounded iterated local mean). *For every degree $d \geq 2$,*

$$\lim_{P \rightarrow \infty} \lim_{K \rightarrow \infty} \lim_{H \rightarrow \infty} \frac{1}{|\mathcal{S}_d(H)|} \sum_{(a,b) \in \mathcal{S}_d(H)} D_{d,n,P,K}(a,b) = L_d(n), \quad (6.13)$$

where, equivalently,

$$L_d(n) = \sum_{j < n} \sum_{k \in \Lambda_{d,j}} \frac{\varphi(k)}{2} \sum_{\substack{p \nmid 2d \\ k|p-1 \text{ or } k|p+1}} \frac{\log p}{p^2 - 1} \quad (6.14)$$

$$= \sum_{j < n} \sum_{m \in \mathcal{L}_{d,j}} \varphi(m) \sum_{\substack{p \nmid 2d \\ \lambda(m)|p-1 \text{ or } \lambda(m)|p+1}} \frac{\log p}{p^2 - 1}. \quad (6.15)$$

There is an explicit constant C_d , independent of n , such that

$$L_d(n) \leq C_d < \infty. \quad (6.16)$$

One may take

$$C_d = 3 \sum_{\substack{m|d^\infty \\ m > 1}} \frac{\varphi(m)}{m^2} (\zeta(2) \log(4m) - \zeta'(2)). \quad (6.17)$$

If

$$B_d = \prod_{q|d} \left(1 + \frac{1}{q}\right),$$

then the same constant has the closed form

$$C_d = 3 \left[(\zeta(2) \log 4 - \zeta'(2))(B_d - 1) + \zeta(2) B_d \sum_{q|d} \frac{q \log q}{q^2 - 1} \right]. \quad (6.18)$$

Proof. Fix $p \nmid 2d$. By Lemma 9 and (6.9), the number of simple projective roots of the j th normalized layer is

$$N_{d,j}(p) = \sum_{\substack{k \in \Lambda_{d,j} \\ k|p-1 \text{ or } k|p+1}} \frac{\varphi(k)}{2}. \quad (6.19)$$

The sum is an exact disjoint union, not an independence assertion. Each root has one lift modulo p^h and $\varphi(p^h)$ unit representatives. Among the $p^{2h} - p^{2h-2}$ primitive pairs modulo p^h , this gives

$$\Pr_{\mathcal{S}_d(H)}(p^h \mid E_{d,j}) \longrightarrow \frac{N_{d,j}(p)}{p^{h-1}(p+1)}. \quad (6.20)$$

The admissibility conditions are at primes dividing $2d$ and separate from this count by the Chinese remainder theorem. Summing (6.20) for $2 \leq h \leq K+1$, followed by the displayed iterated limits, proves (6.13)–(6.14); (6.12) gives (6.15).

For every $q \geq 3$, with the excluded case $p = 2$ omitted, enlarging the eligible primes to the integers $aq \pm 1$ gives

$$\sum_{q|p-1 \text{ or } q|p+1} \frac{\log p}{p^2 - 1} \leq \frac{3}{q^2} (\zeta(2) \log(2q) - \zeta'(2)). \quad (6.21)$$

Indeed the two denominators are bounded below by a^2q^2 and $a^2q^2/2$, respectively, and $\log(aq \pm 1) \leq \log(2aq)$. Since $m \leq \lambda(m) \leq 2m$, summing (6.21) over every nontrivial index supported on the prime divisors of d proves (6.17). Finally,

$$\sum_{m|d^\infty} \frac{\varphi(m)}{m^2} = B_d, \quad \sum_{m|d^\infty} \frac{\varphi(m) \log m}{m^2} = B_d \sum_{q|d} \frac{q \log q}{q^2 - 1},$$

which follows from the finite Euler product and its logarithmic derivative. This proves (6.18) and convergence. $\square$

The theorem deliberately uses the order of limits displayed in (6.13). Passing from this local model to the untruncated integer-box mean would require uniform control of primes growing with the box:

$$\lim_{P \rightarrow \infty} \limsup_{H \rightarrow \infty} \mathbb{E}_{\mathcal{S}_d(H)} [\log W_{d,n} - D_{d,n,P,\infty}] = 0. \quad (6.22)$$

This is a large-square-divisor problem for binary forms. Fixed-modulus Hensel densities do not prove (6.22), and general multivariable results that assume abc cannot be used here without circularity.

What remains for one fixed orbit

Restrict now to the quadratic orbit or one admissible odd-prime-degree orbit. Write $E_j = E_{d,j}$,

$$\delta_j = \log \frac{E_j}{\text{rad}(E_j)}, \quad q_j = \begin{cases} 2^{j+2}, & d = 2, \\ d^{j+1}, & d \text{ odd}. \end{cases} \quad (7.1)$$

Every sufficiently large prime $p \mid E_j$ satisfies

$$p \equiv \pm 1 \pmod{q_j}. \quad (7.2)$$

Let

$$\text{Sq}(N) = \prod_p p^{\lfloor v_p(N)/2 \rfloor}.$$

Nonnegativity and

$$\log \text{Sq}(E_j) \leq \delta_j \leq 2 \log \text{Sq}(E_j)$$

give the exact equivalence

$$\log W_{d,n} = o(d^n) \iff \delta_j = o(d^j) \iff \log \text{Sq}(E_j) = o(d^j). \quad (7.3)$$

Thus the fixed-orbit problem is a subpower bound for the largest square dividing one sparse sequence of cyclotomic values.

There is one unconditional growing window. Stewart's valuation lemma for a fixed Lucas pair implies, outside an effective finite set,

$$\text{ord}_{\mathfrak{p}}(u^m - 1) < p \exp \left(-\frac{\log p}{51.9 \log \log p} \right) \log |\alpha| \log m, \quad (7.4)$$

where $u = \alpha/\beta$.

Theorem 11 (A growing harmless window). *Put $L_j = \log q_j / \log \log q_j$. For every fixed $0 < \gamma < 1/103.8$,*

$$\sum_{\substack{p \leq q_j \exp(\gamma L_j) \\ p^2 | E_j}} (v_p(E_j) - 1) \log p = o(q_j). \quad (7.5)$$

Proof. Choose a' with $2\gamma < a' < 1/51.9$. Since every candidate prime is at least $q_j - 1$, (7.4) contributes a factor at most $e^{-a' L_j}$. There are at most $2(e^{\gamma L_j} + 1)$ candidate integers in (7.2). Multiplying the number of candidates, the window endpoint, the valuation bound, and $\log p$ gives

$$q_j \exp((2\gamma - a' + o(1))L_j) = o(q_j).$$

□

This is larger than every fixed multiple of q_j , but smaller than $q_j^{1+\varepsilon}$ for each fixed $\varepsilon > 0$. Inside such a polynomial window, the exact remaining obstruction can be stated elementarily. For $Y \geq q_j$, put

$$D_j(Y) = \sum_{\substack{p \leq Y \\ p^2 | E_j}} (v_p(E_j) - 1) \log p.$$

For an integer $K \geq 1$, split it as

$$\begin{aligned} T_{j,K}(Y) &= \sum_{\substack{p \leq Y \\ p^2 | E_j}} \min\{v_p(E_j) - 1, K\} \log p, \\ R_{j,K}(Y) &= \sum_{\substack{p \leq Y \\ p^2 | E_j}} (v_p(E_j) - 1 - K)_+ \log p. \end{aligned} \quad (7.6)$$

Theorem 12 (Polynomial window equals its deep tail). *Let $1 \leq \beta < 2$, $Y = q_j^\beta$, and*

$$K_j = o\left(\frac{q_j^{2-\beta}}{\log q_j}\right).$$

Then

$$T_{j,K_j}(Y) = o(q_j), \quad D_j(Y) = o(q_j) \iff R_{j,K_j}(Y) = o(q_j). \quad (7.7)$$

Proof. By (7.2), at most $2(Y/q_j + 1)$ candidate integers occur. Therefore

$$T_{j,K_j}(Y) \leq 2K_j(Y/q_j + 1) \log Y = o(q_j).$$

The equivalence follows from the nonnegative exact split (7.6). □

In particular, squares, cubes, every fixed valuation depth, and depths growing almost as quickly as $q_j^{2-\beta} / \log q_j$ are harmless. The problem is not merely to show that Lucas–Wieferich primes are rare; it is to rule out an aggregate of extraordinarily deep lifts.

This phrasing is exact. Away from the seed and ramified primes, $p^2 \mid E_j$ forces

$$v_p(U_{p-(\Delta/p)}) = v_p(E_j) \geq 2. \quad (7.8)$$

Thus every squared layer prime is Lucas–Wieferich for the fixed pair. If its valuation is at least 3, the corresponding prime ideal is super-Wieferich. Finiteness of those super-Wieferich prime ideals would, for example, imply $D_j(q_j^\beta) = o(q_j)$ for every fixed $\beta < 2$. Fellini and Murty treat such finiteness as an expected hypothesis for integral admissible bases. Our u is generally only an S -unit, and the implication here is the elementary local argument above, not an application of their theorem. The finiteness hypothesis is not presently available.

Deeper finite computation

Finite computation neither proves nor predicts the fixed-orbit asymptotic. It does give exact regression targets for the layer formulas and the Lucas–Wieferich bridge.

The 122-digit quadratic layer

For the quadratic seed $(a_0, b_0, c_0) = (1, 8, 9)$, the level-seven normalized layer is the 122-digit integer

$$E_7 = 7899993675270888986790467637030360278248237512403075459722095 \\ 7868119418164413649973647970771585795292694673449586011206657.$$

GMP-ECM 7.0.7 split the formerly unresolved 98-digit residual by one Pollard $p - 1$ step and two ECM discoveries. Exact division gives

$$E_7 = 189439 \cdot 750692351 \cdot 9825841153 \cdot 298196593663 \\ \cdot 991245449894911 \cdot 6726631000961507661177857 \\ \cdot 28434404151626641091139435909034910237447173121. \quad (8.1)$$

All seven factors have recursive Pocklington certificates based on complete factorizations of $p - 1$. A standard-library verifier checks every modular witness and exact multiplication; the layer is also recomputed from the recurrence. Each factor occurs to exponent one, and their residues modulo 512 are

$$511, 511, 1, 511, 511, 1, 1.$$

Thus the forced $p \equiv \pm 1 \pmod{512}$ condition holds and this layer has zero repeated-prime defect.

A census beyond one million

For a Lucas pair (P, Q) , the census tests every odd unramified prime $p \leq 10^8$ for

$$p^2 \mid U_{p-(\Delta/p)}(P, Q)$$

and then tests the hits modulo p^3 . Two implementations were used: binary exponentiation in $\mathbb{Z}[X]/(X^2 - PX + Q)$, and an independently written companion-matrix algorithm. Each scanned all 5,761,454 odd primes through 10^8 , and the outputs agree exactly.

orbit pair (P, Q)	Wieferich p	rank	tower rank?	depth ≥ 3 ?
quadratic (14, 81)	65519	455	no	no
cubic $(-2, 25)$	47	24	no	no
cubic $(-2, 25)$	31220573	7805143	no	no
quintic $(-6, 49)$	53	26	no	no

The larger search finds the new cubic hit $p = 31220573$, with $\chi = +1$, rank

$$7805143 = 19 \cdot 547 \cdot 751,$$

and exact depth two. None of the four hits can enter the corresponding displayed orbit tower because no rank is a pure power of the orbit degree, and none is super-Wieferich. This is a bounded observation only; no independence or density assertion is being made.

The exact inputs, intermediate cofactors, software versions, machine-readable census output, and regression commands accompany the source repository.

Conclusions

The all-degree results are structural. They show that the quadratic orbit, the odd-prime construction, and every composite block share one normalized Lucas-layer mechanism, one radical identity, and one bounded iterated local mean. The quadratic realization and the all-index atom classification complete two genuine gaps between the first two manuscripts.

The fixed-orbit result is more modest. Congruence and valuation theory control a growing $q_j^{1+o(1)}$ window, and pure counting disposes of every bounded or moderately growing lift depth in any fixed polynomial window below q_j^2 . What survives is a deep diagonal Lucas–Wieferich tail. Neither the complete level-seven factorization nor the census through 10^8 changes that logical boundary.

Accordingly, these papers have sharpened the exact reduction and the partial bounds toward the fixed-orbit question; they have not proved a new case of *abc*. A next theorem would need genuine uniform control of deep lifts for one fixed Lucas pair, or a different global mechanism for the powerful part.

AI-use statement

OpenAI Codex and Anthropic Claude were used during Phases 5–9 for mathematical exploration, proof and source audits, independent software implementations, computation, drafting, and adversarial cross-review. The accompanying Lean project machine-checks the generic prime-element multiplicity identity behind W1 and the Cayley inversion-fiber and primitive-index lemmas used in Lemma 9; it does not formalize the whole paper. The named author must independently verify every theorem, proof, reference, novelty statement, and computation before circulation or submission. The AI systems are not authors.

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